

Fatemeh azam

Tulsa, OK | Fatemeh-azam@utulsa.edu | 4803599428 | [\[LinkedIn\]](#) [\[Portfolio\]](#) [\[GitHub\]](#)

Summary

Software engineer with experience designing and delivering backend and full-stack systems that support large-scale, data-driven production applications. Strong background in Python-based services, RESTful APIs, databases, and performance-sensitive, real-time systems, with a track record of translating evolving, ambiguous requirements into reliable, production-ready solutions. Comfortable working across cross-functional teams, mentoring peers, and driving reliability improvements through metrics analysis. Authorized to work permanently in the U.S. (Green Card) — no employer sponsorship required.

Core Skills

Backend & API Engineering: Python, RESTful API design, Microservices concepts, authentication, validation, performance tuning

Full-Stack Development: React, Node.js, Socket.io, HTML5/CSS3, responsive UI development

Data & Systems: MongoDB, SQL (schema design, indexing, query optimization), SQL Server, real-time data pipelines

DevOps & CI/CD: Git, GitHub Actions, Azure DevOps, unit/integration/API testing

Reliability & Delivery: Production system monitoring, cross-functional delivery, mentoring, AI-assisted development (Copilot, Cursor)

Experience

Software Engineer — Oklahoma Cyber Innovation Institute, University of Tulsa

Tulsa, OK | Oct 2024 – Present

- Developed Python-based backend services and RESTful APIs powering interactive, data-driven production applications, integrated with React/Node.js front-ends
- Designed and integrated MongoDB and SQL data pipelines across a distributed multi-node sensor network, supporting real-time analytics, reporting, and reliable system behavior
- Collaborated with cross-functional research, hardware, and deployment teams to translate evolving, ambiguous requirements into maintainable, production-ready solutions
- Analyzed system metrics and performance data to drive reliability improvements, supporting live production deployments and public technical demonstrations
- Partnered with security-focused efforts to test systems against real-world failure and attack scenarios, building secure-by-design practices into development early
- Mentored junior contributors through code review and technical guidance, and presented deployed systems to technical and non-technical audiences at statewide innovation showcases

Project / Event Coordinator — University of California San Diego

San Diego, CA | Nov 2022 – Mar 2024

- Led cross-functional teams and developed streamlined project planning processes, increasing delivery efficiency
- Recognized by supervisor for contributing innovative process strategies

Software Engineer — IRISA Co.

Isfahan, Iran | 2010 – 2016

- Built high-throughput backend modules for log ingestion and processing in Python and C++ using multithreading/multiprocessing
- Designed data pipelines and structured storage workflows supporting real-time analysis at scale
- Developed and integrated anomaly detection and analytics components into production environments, focused on performance, accuracy, and reliability
- Optimized queries and processing for large volumes of system/network events

Education & Publications

M.S. in Software Engineering — University of Isfahan, Iran | 2008–2011

B.S. in Software Engineering — University of Isfahan, Iran | 2003–2008

Accepted paper, "Authority Contention in Micro-UAV Command Stacks" — IEEE CARS 2026 (Autonomous Systems / CAE-R track)